

Frozen Baseline Review Report

Round 14 packaging pass

1 Executive Summary

This report freezes the pre-improvement baseline before any future algorithm-strengthening work. The current benchmark inventory contains 14 packaged experiment runs: 3 exact, 6 epsilon-certified, 5 smoke-only, and 0 failed. The validation chain from reference disaster primal through the production Benders engine is already built and individually checked, but the harder exact-mode convergence cases remain non-certified.

- Exact results currently cover the fixed disaster primal/dual reference chain, the normal-only direct master baselines, and selected tiny or longer-budget exact-mode Benders diagnostics.
- Epsilon-certified results are bounded certificates relative to their chosen tolerance; they are not exact-zero convergence.
- Smoke-only runs remain visible and are not promoted into certified evidence.
- Default `runtime_12` remains a local directional benchmark and is not numerically paper-Table-II comparable.

2 Current Frozen Contract

- Source-of-truth hierarchy: frozen specs under `docs/spec/`, validated round reports under `docs/reports/`, human-facing interpretation packs under `docs/analysis_packs/`, and current packaged outputs under `results/`.
- Explicit critical-bus freeze for this report: [2, 5, 8, 11, 14, 17, 20, 23, 26, 29, 32].
- `smoke_only` means a bounded directional run that is not certified evidence.
- `epsilon_certified` means the run stopped with a proved violation upper bound below the configured epsilon, not exact zero.
- The paper-like family is an approximation layer built from local config overrides, not paper-number reproduction.

3 Validation Chain Summary

4 Benchmark Experiment Summary

Certified-small family (regenerated in this round because the Round 12 packaged outputs no longer carried those rows separately):

Table 1: Validation chain summary for the frozen pre-improvement baseline.

Round	Validated layer	Status	Key result
02	reference disaster primal	exact toy + runtime fixture	Eq. (26)-(32) fixed-(x,delta,b) primal
03	canonicalizer + auto dual	exact objective/KKT matching	mechanical canonicalization and auto-
04	paper dual	exact samplewise decomposition	beta/gamma/phi decomposition match
05	separation oracles	exact tiny validation	outage enumeration, budget support, c
05.5	separation hardening	coefficient/model-shape validation	runtime bounds and edge-case checks
06	first-stage + normal block	toy-case + runtime-smoke validated	Eq. (10)-(25) production blocks built
06.5	unit/interface hardening	analytic Eq. (24) guard	voltage-drop scaling and first-stage AF
07	fixed-cut master	toy-case + runtime-smoke validated	restricted master for Eq. (33), (37)-(39)
07.5	master/cut interface	real-cut interface hardened	paper-dual decomposition enters master
08	single generated cut step	one-step efficacy validated	master -i separation -i cut -i master p
09	full Benders engine	tiny exact + runtime smoke	iterative Benders loop validated agains
10	independent disaster exact oracle	tiny cross-check + readiness	primal-exact disaster oracle agrees wit
11	experiment interpretation pack	packaging only	benchmark packaging and explicit vali
12	paper-style experiment pack	packaging only	paper-like and runtime-directional fam
12.5	convergence diagnostics	diagnostic only	exact_zero / epsilon_stop / max_iter_n
13	cut-process diagnostics	diagnostic only	repeated outages observed; exact dupl

- integrated_mainline_certified_small: validation=epsilon_certified, stop=certified_epsilon, construction=4,421,143.360, weighted normal=20,588,064.250, disaster master=167.617432
- normal_only_certified_small: validation=exact, stop=direct_optimal, construction=4,421,143.360, weighted normal=22,875,626.944, disaster master=0.000000
- deterministic_mean_value_certified_small: validation=epsilon_certified, stop=certified_epsilon, construction=4,421,143.360, weighted normal=20,588,064.250, disaster master=167.617432

Paper-like family:

- integrated_mainline_paper_like: validation=epsilon_certified, stop=certified_epsilon, construction=171,730.136, weighted normal=13,035,357.592, disaster master=0.000000
- normal_only_paper_like: validation=exact, stop=direct_optimal, construction=171,760.064, weighted normal=18,622,035.264, disaster master=0.000000
- disaster_only_paper_like: validation=smoke_only, stop=max_iterations, construction=23,113.825, weighted normal=0.000000, disaster master=3,709.868
- deterministic_mean_value_paper_like: validation=epsilon_certified, stop=certified_epsilon, construction=171,730.136, weighted normal=13,035,357.592, disaster master=0.000000
- ev_penetration_1.5x_paper_like: validation=epsilon_certified, stop=certified_epsilon, construction=262,853.694, weighted normal=17,010,934.514, disaster master=0.000000
- ev_penetration_2.0x_paper_like: validation=epsilon_certified, stop=certified_epsilon, construction=305,364.115, weighted normal=25,867,747.207, disaster master=0.000000

Runtime12 directional family:

- integrated_mainline_runtime12: validation=smoke_only, stop=max_iterations, construction=4,410,273.924, weighted normal=20,506,437.556, disaster master=87.352783

- normal_only_runtime12: validation=exact, stop=direct_optimal, construction=4,410,273.924, weighted normal=22,784,930.618, disaster master=0.000000
- deterministic_mean_value_runtime12: validation=smoke_only, stop=max_iterations, construction=4,183,374.434, weighted normal=20,674,646.361, disaster master=103.570653
- ev_penetration_1_5x_runtime12: validation=smoke_only, stop=max_iterations, construction=6,172,481.342, weighted normal=30,577,949.733, disaster master=122.108885
- ev_penetration_2_0x_runtime12: validation=smoke_only, stop=max_iterations, construction=6,470,032.170, weighted normal=46,930,921.151, disaster master=122.108885

Validation labels by packaged experiment run:

- - ‘exact’: ‘normal_only_certified_small’, ‘normal_only_paper_like’, ‘normal_only_runtime12’
- - ‘epsilon_certified’: ‘integrated_mainline_certified_small’, ‘deterministic_mean_value_certified_small’, ‘integrated_mainline_paper_like’, ‘deterministic_mean_value_paper_like’, ‘ev_penetration_1_5x_paper_like’, ‘ev_penetration_2_0x_paper_like’
- - ‘smoke_only’: ‘disaster_only_paper_like’, ‘integrated_mainline_runtime12’, ‘deterministic_mean_value_runtime12’, ‘ev_penetration_1_5x_runtime12’, ‘ev_penetration_2_0x_runtime12’
- - ‘failed’: none

5 IEEE 33-Bus Station Topology Interpretation

Figure 1 shows the runtime12 directional maps, while Figures 2 and 3 show the paper-like maps. Critical buses are highlighted explicitly in red. The current baseline supports a careful topological interpretation: integrated and normal-only plans often place direct capacity at or near critical buses, fast-charger hubs concentrate on a smaller subset of opened buses, and EV-penetration expansion is monotone in buildout. None of these plots should be read as paper-number reproduction.

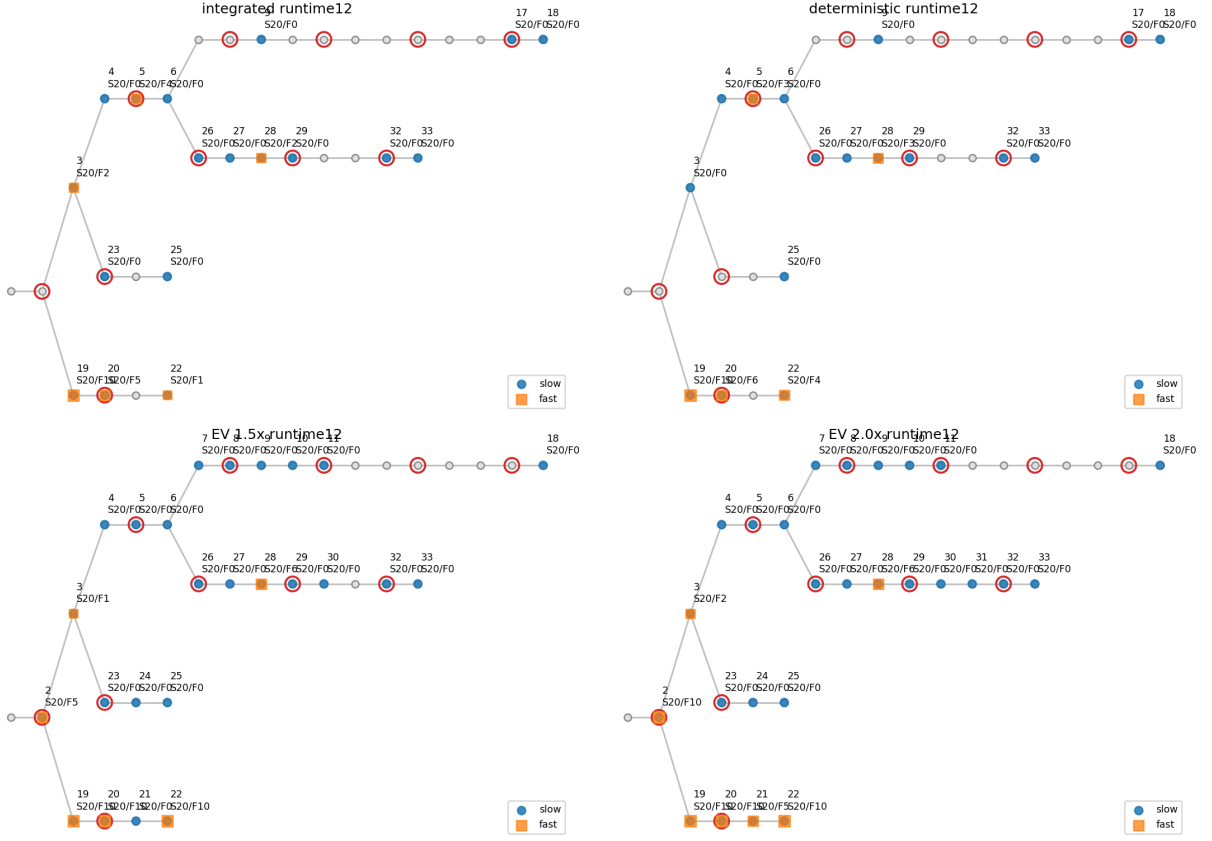


Figure 1: Runtime12 directional topology maps.

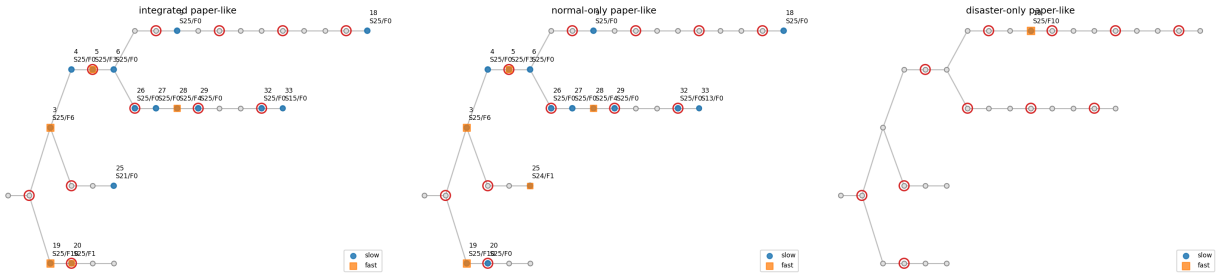


Figure 2: Paper-like case 1/2/3 topology maps: integrated, normal-only, disaster-only.

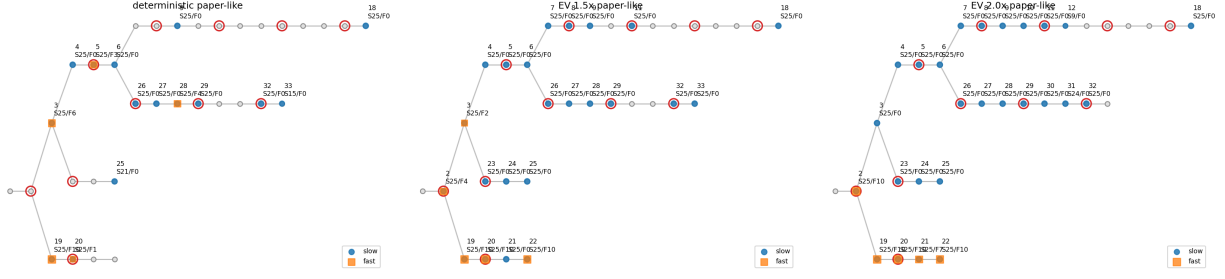


Figure 3: Paper-like case 4/5/6 topology maps: deterministic mean-value and EV-penetration cases.

6 Objective Decomposition

Figure 4 uses the weighted normal term consistently and keeps the unweighted normal term out of the main stacked chart. Cross-family ranking by raw total objective is intentionally avoided in the safer benchmark comparison view of Figure 5.

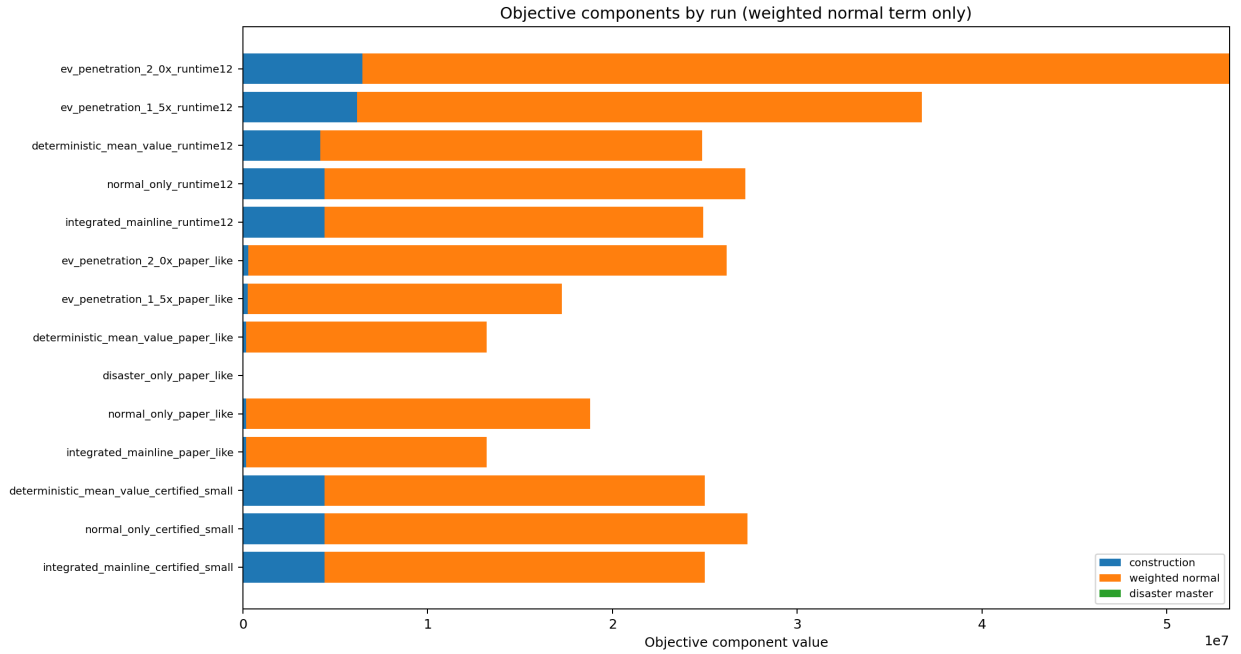


Figure 4: Construction, weighted normal term, and disaster master term by run.

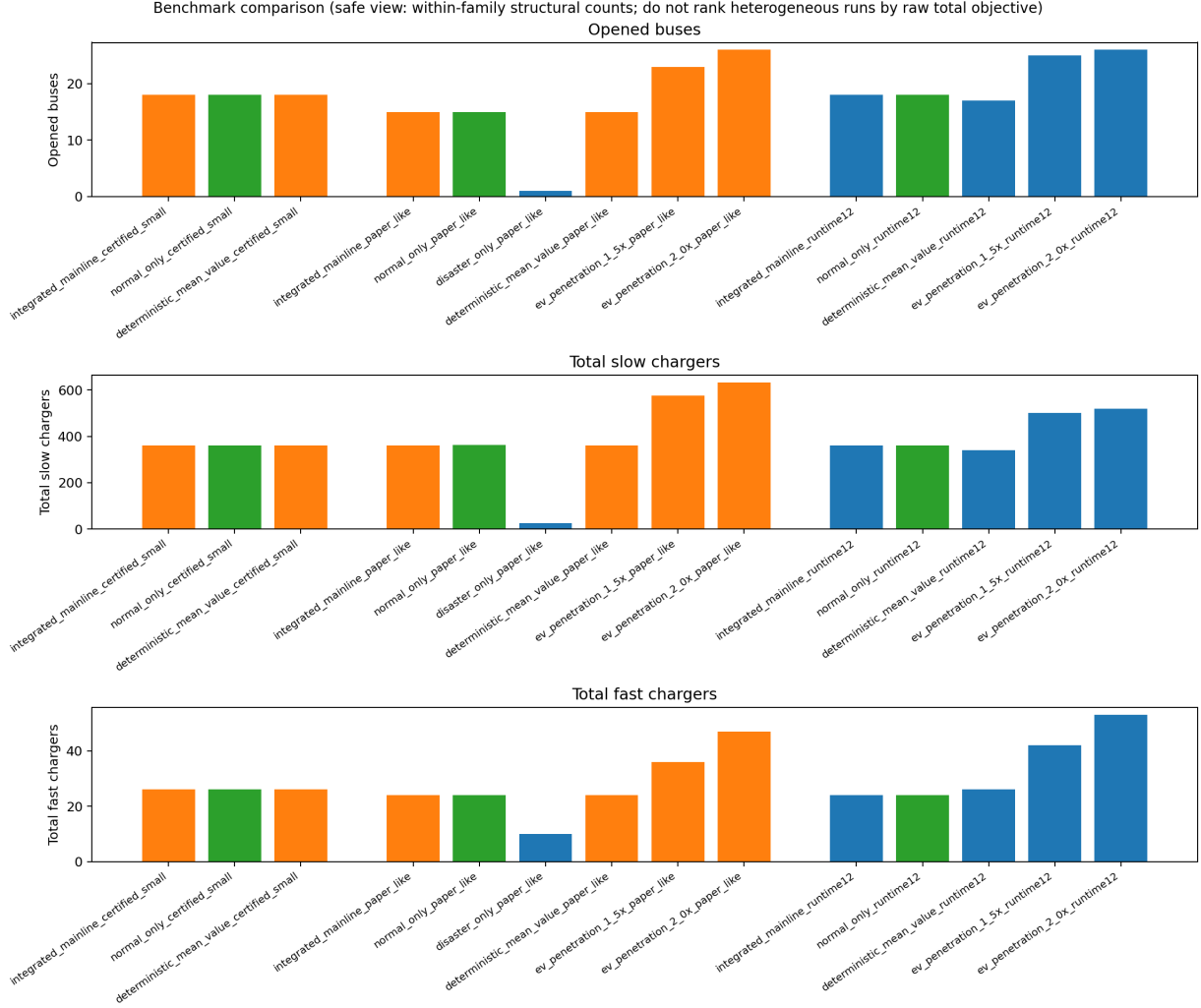


Figure 5: Safer benchmark comparison based on structural counts within each family.

7 Convergence / Non-Certification Diagnostics

The convergence and cut-process diagnostics keep non-certified runs visible. Round 12.5 showed that exact-zero, epsilon-stop, and max-iteration non-certification need to be separated explicitly. Round 13 showed repeated outages in several hard cases, while exact duplicate structured cut signatures were not observed.

- Repeated-outage cases from Round 12.5: integrated_paper_like_exact_i20_k2_a1b1, disaster_only_paper_like_exact_i20_k2_a1b1.
- Repeated structured-cut signature cases from Round 12.5: none observed.
- Hard exact-mode or strengthened diagnostic cases still non-certified: integrated_paper_like_exact_i10_k2_a1b1, integrated_paper_like_exact_i20_k2_a1b1__baseline, disaster_only_paper_like_exact_i5_k1_a1b1__baseline, disaster_only_paper_like_exact_i20_k1_a1b1__baseline, disaster_only_paper_like_exact_i5_k2_a1b1__baseline, disaster_only_paper_like_exact_i20_k2_a1b1__baseline, disaster_only_paper_like_exact_i40_k2_a1b1__baseline, integrated_runtime12_exact_i3_k3_a1234b1234__baseline.

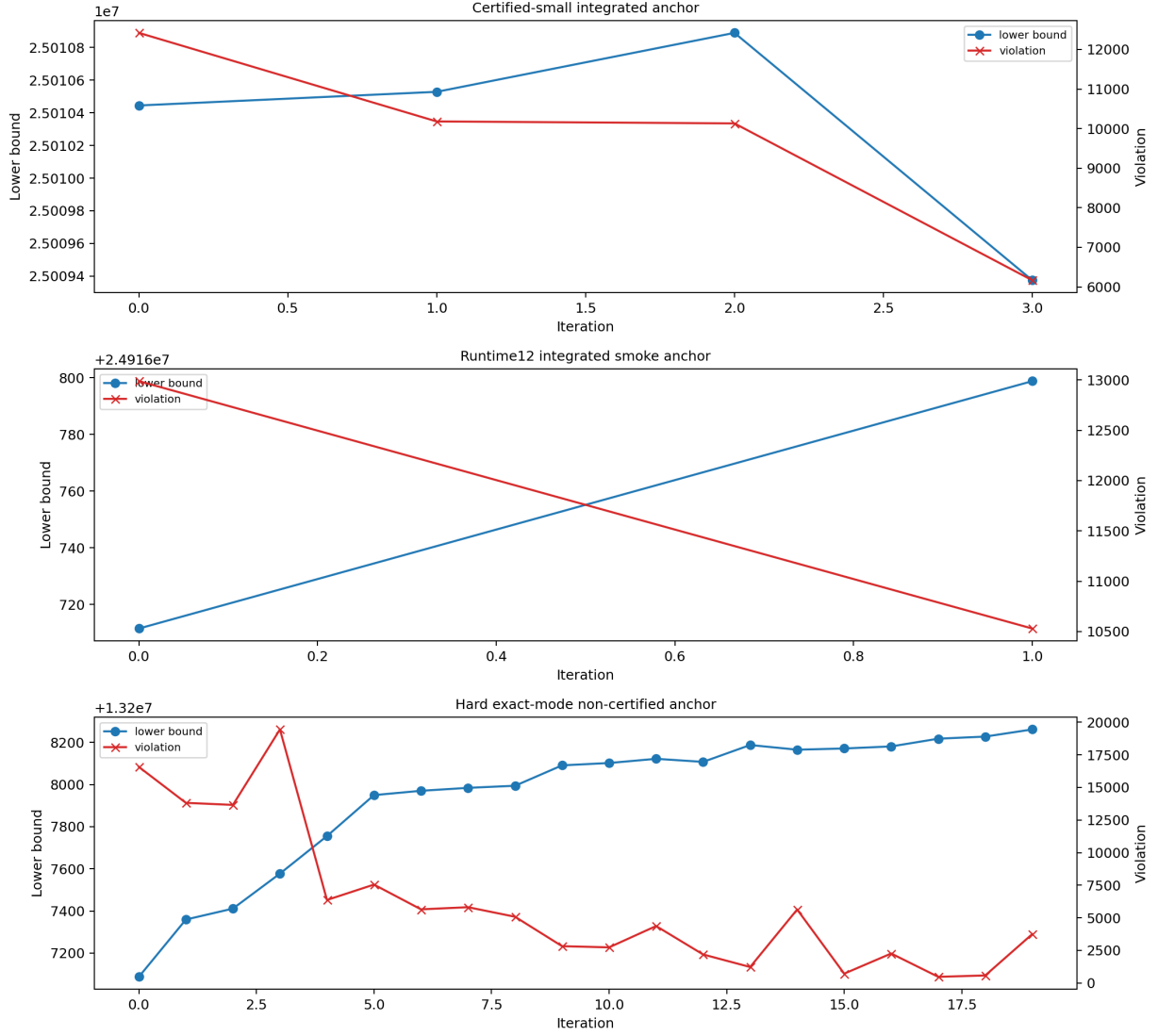


Figure 6: Selected iteration traces: certified-small anchor, runtime smoke anchor, and hard exact-mode non-certified anchor.

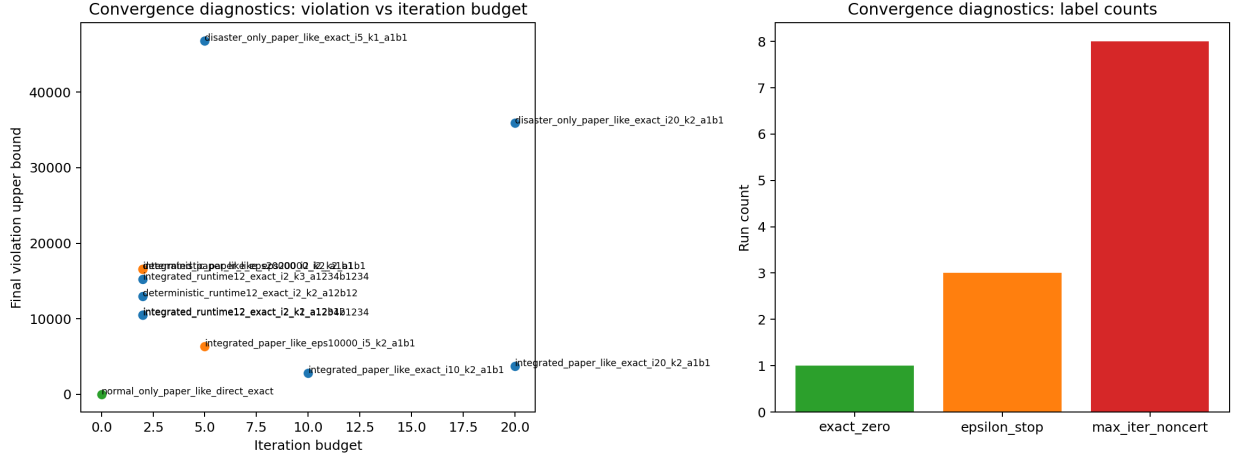


Figure 7: Round 12.5 convergence-diagnostic overview.

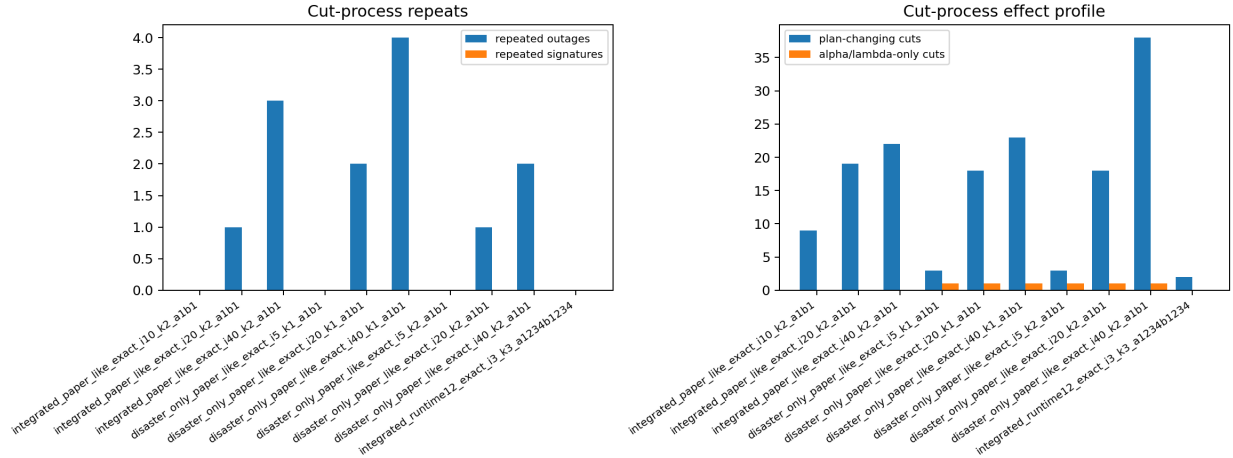


Figure 8: Round 13 cut-process overview.

8 Safe Claims vs Unsafe Claims

Safe now.

- The reference-to-production validation chain has been individually exercised from the disaster primal through the fixed-cut master and iterative Benders engine.
- Exact, epsilon-certified, smoke-only, and failed labels are preserved explicitly in the frozen baseline.
- The current diagnosis points at cut-process effectiveness as the main remaining difficulty in the harder exact-mode runs, not at raw-data loading or a known core-math break.

Not safe yet.

- Default `runtime12` is not paper-Table-II comparable.

- The paper-like family is not a paper-number reproduction.
- Heterogeneous benchmark runs should not be ranked by raw total objective without caveats.
- Hard non-certified exact-mode runs should not be described as converged.

9 Next Improvement Priorities

1. Strengthen cut-process effectiveness rather than relabeling smoke-only outcomes.
2. Revisit multi-cut or stabilization ideas only after preserving the current exact/epsilon/smoke discipline.
3. Keep baseline archive, anchor reruns, and honest non-certification visibility intact during future improvement rounds.

Anchor Consistency Checks

The round reran three anchors against archived/current baselines:

Anchor	Run id	Objective match	Validation match	Stop match	Notes
certified_small_integrated	certified_mainline_certified_small	True	True	True	baseline source was re-generated because current Round 12 outputs no longer include certified-small artifacts
runtime12_integrated	runtime12_mainline_runtime12	True	True	True	rerun matched baseline within configured tolerances
hard_exact_mode	integrated_paper_like_exact_i20_k2_alpha1	True	True	True	rerun matched baseline within configured tolerances